

Choosing S

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To use the marker we want to find an single-particle operator S , that is odd with respect to time-reversal invariance, $\mathcal{T}S\mathcal{T}^{-1} = -S$, squares to unity and is a local tensor sum

$$S = \bigoplus_r s_r. \quad (1)$$

Here r can label any unit-cell, and the local Hilbert-space dimension d_r can in principle vary across the system. The difficulty is making such a choice and at the same time

$$\tilde{S} = \{\rho, S\} - S \quad (2)$$

which by construction has eigenvalues between -1 and 1 , have a large spectral gap around 0 .

1 The worst case scenario

We will go through quite generic situations where one can avoid the hassel mentioned in this section, but lets first go through the worst case scenario.

In the most difficult case we really have to optimize the band gap of $\tilde{S}$ over the space of choices for S . Lets outline how this is done here. We now optimize these vectors such that $\tilde{S}^2$ is as close to the identity as possible, in particular avoiding eigenvalues near zero. A natural choice is the cost function $CF = -\log \det \tilde{S}^2$, which is a smooth, basis-independent function that penalizes each eigenvalue individually. Since the eigenvalues of $\tilde{S}^2$ are in the range $[0, 1]$, the cost is uniquely minimized when all eigenvalues are equal to unity, while diverging as any eigenvalue approaches zero. This provides a robust objective that simultaneously favors $\tilde{S}^2$ being unity and strongly suppresses nearly singular solutions. Moreover, as a function of the positive-definite matrix $\tilde{S}^2$, it is convex and self-concordant, giving it favorable numerical properties as an optimization objective.

Using the fact that S squares to unity and that ρ is a projector we get $\tilde{S}^2 = 1 + [S, \rho]^2$ and

$$CF(S) = -\text{Tr} \log(1 + [S, \rho]^2) \quad (3)$$

1.1 Hessians and gradients for the untruncated const-function

To do the optimization efficiently we need to introduce an inner product and find the Hessian and gradient, lets first find the Hessian and gradient with respect to S . For the space of Hermitian matrices the canonical inner product is the Hilbert-Schmidt product, i.e., the product of two matrices A and B is defined as $\text{Tr}(AB)$. We write

$$CF[S + \delta_S] = CF[S] + \text{Tr}(\delta_S \nabla_S CF) + \frac{1}{2} \text{Tr}(\delta \mathcal{H}_S^{CF}(\delta)) + \mathcal{O}(\delta^3) \quad (4)$$

where $\mathcal{H}_S^{CF}$ is a linear (super)operator symmetric w.r.t. the Hilbert-Schmidt inner product, i.e., for hermitian matrices $\text{Tr}(A\mathcal{H}_S(B)) = \text{Tr}(B\mathcal{H}_S(A))$. But lets do it stepwise, first w.r.t. $\tilde{S}^2$. Using the fact that the determinant of a product is the product of determinants we get $\det(\tilde{S}^2 + \delta_{\tilde{S}^2}) = \det(\tilde{S}^2) \det(1 + \tilde{S}^{-2} \delta_{\tilde{S}^2})$ and thus $CF[\tilde{S}^2 + \delta_{\tilde{S}^2}] = CF[\tilde{S}^2] - \text{Tr} \log(1 + \tilde{S}^{-2} \delta_{\tilde{S}^2})$, and

$$\log(1 + \tilde{S}^{-2} \delta_{\tilde{S}^2}) = \tilde{S}^{-2} \delta_{\tilde{S}^2} + \frac{1}{2} \tilde{S}^{-2} \delta_{\tilde{S}^2} \tilde{S}^{-2} \delta_{\tilde{S}^2} + \mathcal{O}(\delta_{\tilde{S}^2}^3). \quad (5)$$

Now

$$\begin{aligned} 1 + [S + \delta, \rho]^2 &= 1 + ([\delta, \rho] + [S, \rho])^2 \\ &= \tilde{S}^2 + \{[S, \rho], [\delta, \rho]\} + [\delta, \rho]^2 \end{aligned} \quad (6)$$

Using the cyclic property of the trace, that $[S, \rho]$ and $\tilde{S}^{-2}$ commute, and $\text{Tr}([A, B]C) = \text{Tr}(A[B, C])$ we can simplify the first order term in δ_S of $CF[S + \delta_S]$ as

$$\text{Tr}(\{[S, \rho], [\delta, \rho]\} \tilde{S}^{-2}) = 2\text{Tr}(\delta[\rho, [S, \rho] \tilde{S}^{-2}]), \quad (7)$$

and we can read off the gradient

$$\nabla_S CF = 2[\rho, [S, \rho] \tilde{S}^{-2}]. \quad (8)$$

Now the second order term is

$$-\text{Tr}([\delta, \rho]^2 \tilde{S}^{-2}) + \frac{1}{2} \text{Tr}(\tilde{S}^{-2} \{[S, \rho], [\delta, \rho]\} \tilde{S}^{-2} \{[S, \rho], [\delta, \rho]\}) \quad (9)$$

The second term

$$\frac{1}{2} \text{Tr}(\tilde{S}^{-2} \{[S, \rho], [\delta, \rho]\} \tilde{S}^{-2} \{[S, \rho], [\delta, \rho]\}) = \quad (10)$$

$$\frac{1}{2} \text{Tr}(\tilde{S}^{-2} \{[\delta, \rho], [S, \rho]^2\} \tilde{S}^{-2} [\delta, \rho]) \quad (11)$$

$$+ \text{Tr}([\delta, \rho] \mathcal{C}_{\tilde{S}^{-2}[S, \rho]}([\delta, \rho])) \quad (12)$$

where $\mathcal{C}_{\mathcal{O}}$ is the conjugation superoperator, i.e., for an Hermitain $\mathcal{O}$ $\mathcal{C}_{\mathcal{O}}(X) = \mathcal{O}X\mathcal{O}$. The first of these can in turn be simplified as

$$\frac{1}{2} \text{Tr}([\delta, \rho] \tilde{S}^{-2} \{[\delta, \rho], [S, \rho]^2\} \tilde{S}^{-2}) = \text{Tr}([\delta, \rho]^2 \tilde{S}^{-2}) - \text{Tr}([\delta, \rho] \mathcal{C}_{\tilde{S}^{-2}}([\delta, \rho])). \quad (13)$$

Using this form and collecting the terms we get

$$-\text{Tr}(\delta i[\rho, (\mathcal{C}_{\tilde{S}^{-2}} - \mathcal{C}_{\tilde{S}^{-2}[S, \rho]})(i[\rho, \delta])). \quad (14)$$

Now we can also read-off the Hessian, but since the terms can be written in different orderings we first have to check that we have written the terms in a symmetric manner. The super-operator $\text{ad}_{\mathcal{O}}(X) = i[\mathcal{O}, X]$ is, for hermitian $\mathcal{O}$, anti-symmetric w.r.t. the Hilbert-Schmidt inner-product, and the superoperator $\mathcal{C}_{\mathcal{O}}$ is symmetric. So the first term is of the form $\mathcal{A}\mathcal{S}\mathcal{A}$ where $(\mathcal{A})\mathcal{S}$ is (anti)-symmetric, so $(\mathcal{A}\mathcal{S}\mathcal{A}) = \mathcal{A}^T \mathcal{S}^T \mathcal{A}^T = \mathcal{A}\mathcal{S}\mathcal{A}$, where T denote Hilbert-Schmidt transpose. So we can conclude

$$\mathcal{H}_S^{CF}(X) = -2i[\rho, (\mathcal{C}_{\tilde{S}^{-2}} + \mathcal{C}_{i[S, \rho]\tilde{S}^{-2}})(i[\rho, X])). \quad (15)$$

If A is anti-symmetric, and B is semi positive-definite then $-ABA$ is semi-positive definite, $-\text{Tr}(XABA(X)) = \text{Tr}(A(X)BA(X)) \geq 0$, and $\mathcal{C}_{\mathcal{O}}$ is also semi-positive definite for any hermitian $\mathcal{O}$ (for hermitian X , $\text{Tr}(X\mathcal{C}_{\mathcal{O}}(X)) = \text{Tr}((X\mathcal{O})^2) \geq 0$) so we also explicitly see that $\mathcal{H}_S^{CF}$ is semi-positive definite.

1.2 Truncated cost-function

We can approximate the cost-function as

$$-\text{Tr} \log(1 + [S, \rho]^2) = ||[S, \rho]||_F^2 + \mathcal{O}([S, \rho]^4). \quad (16)$$

Using $S^2 = 1$ and $\rho^2 = \rho$ and assuming half filling $\text{Tr} \rho = D/2$, we have

$$||[S, \rho]||_F^2 = D-2\text{Tr} S \mathcal{C}_{\rho}(S) \quad (17)$$

1.3 Translation invariant ansatz

1.3.1 Parametrizing s

We now move on to explicitly parametrize S such that it is odd under time-reversal and squares to unity. I will first specialize to translation invariant S , i.e., $s_r = s$, and use local Hilbert-space dimension $d = 4$. Let $\{\tau_i\}$ and $\{\sigma_i\}$ be two mutually commuting sets of Pauli-operators on the local Hilbert-space and they can be chosen such that time-reversal takes the form $\mathcal{T} = i\sigma_2 K$.

A general hermitian operator can be written as

$$s = \sum_{\mu\nu=0}^3 c_{\mu\nu} \sigma_{\mu} \tau_{\nu}. \quad (18)$$

Since the only TRI odd τ is τ_2 and all but σ_0 are TRI odd, a general TRI odd operator is of the form

$$s = n_0 \cdot \vec{\sigma} + n_1 \cdot \sigma \tau_1 + n_3 \cdot \sigma \tau_3 + m \tau_2 \quad (19)$$

Now lets enforce, $s^2 = 1$. We have

$$s^2 = (|n_0|^2 + |n_1|^2 + |n_3|^2 + m^2) + 2(n_0 \cdot n_1)\tau_1 + 2(n_0 \cdot n_3)\tau_3 + 2(n_1 \times n_3 - mn_0) \cdot \sigma \tau_2. \quad (20)$$

and thus $n_0 = n_1 \times n_3/m$ and we have the parametrization

$$s = \frac{(n_1 \times n_3)}{m} \cdot \vec{\sigma} + (n_1 \cdot \sigma)\tau_1 + (n_3 \cdot \sigma)\tau_3 - m\tau_2 \quad (21)$$

with the rather inconvenient normalization

$$F(n_1, n_3, m) = \frac{|n_1|^2 |n_3|^2 - (n_1 \cdot n_3)^2}{m^2} + (|n_1|^2 + |n_3|^2) + m^2 = 1. \quad (22)$$

This is of course singular when m is small, so if the optimization flows towards small m it is probably good to first do the an optimization with $m = 0$ and then improve around that (if needed). For $m = 0$ we can write

$$s = n \cdot \vec{\sigma} + (\tilde{n} \cdot \sigma)(n_0 \cdot \tau) \quad (23)$$

with $|n_0| = 1$ and having only components in the 1 and 3 direction, $n \perp \tilde{n}$ and $|n|^2 + |\tilde{n}|^2 = 1$. Since $n \perp \tilde{n}$ it is convenient to switch to $n_1 = \frac{1}{\sqrt{2}}(n + \tilde{n})$ and $n_2 = \frac{1}{\sqrt{2}}(n - \tilde{n})$ in those variables the only constraint is the norm, i.e., n_0, n_1, n_2 are all unit vectors.

1.3.2 The truncated cost-function

Evaluating (17) with $m_\alpha = (\vec{\sigma}, \vec{\sigma}\tau_1, \vec{\sigma}\tau_3, \tau_2)$ i.e.,

$$A_{\alpha\beta} = -2\text{Tr}m_\beta C_\rho(m_\alpha) \quad (24)$$

we get a 10by10 matrix A and evaluating the truncated costfunction just amounts to calculating the mean under that matrix. So when the matrix is precalculated the evaluation is extremely fast. This means that one can just plug this form together with the constraint $F = 1$ into built in minimum finders for some randomly chooses intial guesses. In mathematica its called FindMinimum. The problem is that the first truncated ie just frobenius norm does not disfavor evs close to 0 enough. One could try to add also next level in truncation,

$$-\text{Tr} \log(1 + [S, \rho]^2) = ||[S, \rho]||_F^2 + \frac{1}{2}\text{Tr}([S, \rho]^4) + \mathcal{O}([S, \rho]^6). \quad (25)$$

Then the problem would no longer just be the expectation value of a matrix of the vector of 10 params but also a 4 index tensor.

1.3.3 Parametrizing the vectors

Since F above is homogeneous it is convenient to write a general parametrization in the following way. At some point (n_1^0, n_3^0, m^0) take the six directions v_i orthogonal to ∇F , and use a_i below as the parameters to parametrize (n_1, n_3, m)

$$(n_1, n_3, m) = \frac{(n_1^0, n_3^0, m^0) + \sum_i a_i v_i}{F((n_1^0, n_3^0, m^0) + \sum_i a_i v_i)}. \quad (26)$$

1.3.4 Optimizing the untruncated

The full const-function is very expensive to calculate, so we need to do some of the heavy-lifting our selves and cant use black-box optimizers if we want the optimization to be fast.

As a function of S the cost-function is convex and self-concordant, meaning that Newtons method would converge extremely fast. I.e., say we are at a parameter x to take a direction v to update in

$$v = -\mathcal{H}^{-1}g \quad (27)$$

where g is the gradient and $\mathcal{H}$ is the Hessian. It is still important not to just take $x \rightarrow x + v$, but to do a line search, i.e., minimize the 1d problem

$$CF(x + (1 + t)v) = \text{const.} + at + bt^2 + \mathcal{O}(t^3) \quad (28)$$

where the coefs a and b are found by series expanding around $x + v$.

Now unfortunately since the cosntraint $s^2 = 1$ is not convex, the problem does not remain convex. With a parametrization where the cosntraint is automatically satisfied to linear order as XXX much is inherited from the nice const-function we started with. One can always try check if the eigenvalues of the Hessian is all positive. If not one can either use stochastic gradient descent, which is slow but easy to implement (and is the one used eg for training neural nets). I.e., just step in the gradient direction but add a small random direction, to avoid getting stuck at a saddle point, and then do line-search in that direction. Doing line-search when the problem is not convex is a bit harder. Then one first can try to expand to second order $CF(x + tv) = \text{const.} + at + bt^2$, if b is positive one checks $t_{min} = -\frac{a}{2b}$ that $\tilde{b}$ in $CF(x + (t_{min} + t)v) = \text{const.} + \tilde{a} + \tilde{b}t^2$ is positive, and in that case the line-search is exactly as in the convex case above. If either of b or $\tilde{b}$ is negative then one have to add higher orders in the initial expansion $CF(x + tv) = \text{const.} + at + bt^2 + ct^3 + \dots$ and minimize that instead.

Since the parameter space is small, one should hopefully fast enter the convex region and then Newtons method will converge in very few steps. If its too slow, one can switch to a second order method, that is more involved, I have done that several times before, so I have some experience about it, but I dont have time to write about it now. But we can talk about it if needed.

1.4 Beyond translation invariance

If we stick to local Hilbert-space dimension $d_r = 4$ the above can be generalized straight forwardly, just by adding a spatial index $s_r = n_r \cdot \vec{\sigma} + (n_r^0 \cdot \vec{\tau})(\tilde{n}_r \cdot \vec{\sigma})$. But it of course become much more involved, since there is then $3N_x N_y$ vectors instead of just 3 that are to be minimized over. We can probably stick with just translation-invarariant ansatz, or at least choose examples where that works. For the full minimization it can in the worst case be quite hard to find the global minima, since there are in general many saddle points.

2 When ρ is strongly localized

In the simplest case when there is a choice of unit-cells such that single particle density matrix ρ is strongly localized i.e., $\{\rho_{rr}\}$ dominate the off-diagonal contributions then we can simply choose S_r to commute with ρ_{rr} and otherwise be random. Lets assume we have chosen the unit-cell small enough such that the only degeneracies of ρ_{rr} is the Kramers degeneracies. The commutation relation $[S_r, \rho_{rr}] = 0$ implies that S_r is diagonal in the same basis as ρ_{rr} ,

$$S_r = \sum_{s_\lambda} s_\lambda |\lambda\rangle \langle \lambda| \quad (29)$$

where $s = \pm 1$ (since S_r should square to unity) and $\{|\lambda\rangle\}_\lambda$ is an orthonormal eigenbasis of ρ_{rr} . But since S_r should be odd under time-reversal $TS_rT^{-1} = -S_r$, while ρ_{rr} is even $T\rho_{rr}T^{-1} = \rho_{rr}$, this means that S_r has to have the form that for Kramers pairs $|\lambda\rangle = T|\bar{\lambda}\rangle$ s must have opposite signs, ($s_{\bar{\lambda}} = -s_\lambda$). Let $\{W_K\}_K$ denote $2 \times d_r$ matrices consisting of the Kramers pairs of ρ_{rr} ,

$$W_K = \begin{pmatrix} \langle \lambda | \\ \langle \lambda | T \end{pmatrix} \quad (30)$$

then we can write S_r as

$$S_r = \sum_K W_{Kr}^T \sigma_3 W_{Kr}^*. \quad (31)$$

Equivalently, the W 's are defined by the eigenvectors of ρ_{rr} as follows. Let $\tilde{U}^*$ denote eigenvectors of ρ_{rr} ordered by magnitude of the eigenvalue. Since we assume that there is no other degeneracy than the Kramers, we can define the odd rows of U as the same as $\tilde{U}$ and the even as the time-reversal partner of the odd,

$$U_{2i-1} = \tilde{U}_{2i-1} \quad ; \quad U_{2i} = T\tilde{U}_{2i-1}. \quad (32)$$

Then we just have

$$W_K = \langle K | \otimes \mathbb{1} U^* \quad (33)$$

where $\langle 1 | = (1, 0, \dots)$ and $\langle 2 | = (0, 1, 0, \dots)$, etc. However, if we multiply each W_K by any $SU(2)$, it is still a Kramers-pair matrix, so (31) is only defined up to $d/2$ $SU(2)$ matrices. Plugging in an arbitrary $SU(2)$ matrix defined by a unit-vector $\hat{n}_{Kr}$ in (31) we get

$$S_r = \sum_K W_{Kr}^T (\hat{n}_{Kr} \cdot \vec{\sigma}) W_{Kr}^*, \quad (34)$$

or equivalently

$$S_r = \sum_{K\alpha} \hat{n}_{rK\alpha} \Sigma_{K\alpha}^r, \quad (35)$$

where

$$\Sigma_{K\alpha}^r = W_{Kr}^T \sigma_\alpha W_{Kr}^*, \quad (36)$$

so $\Sigma_{K\alpha}^r$ is a vector of $m = 3D/2$ $d \times d$ matrices. The n 's can then simply be chooses randomly or in any other fashion.